Set_A_Lab8 - Results

Attempt 1 of 1

Written Mar 11, 2026 7:17 PM - Mar 11, 2026 7:26 PM

Attempt Feedback

15/15
30% penalty
Absent March 11

Attempt Score	**10.5 / 15 - 70 %**
Overall Grade (Highest Attempt)	**10.5 / 15 - 70 %**

Question 1

Use the online https://www.onlinegdb.com/online_c_compiler then run the following code.

```c
#include <stdio.h>
#include <pthread.h>
#include <stdatomic.h>

volatile atomic_int lock = 0;
int counter = 0;

int TSL(volatile atomic_int *lock) {
    return atomic_exchange(lock, 1);
}
void acquire_lock(volatile atomic_int *lock) {
    while (TSL(lock));
}
void release_lock(volatile atomic_int *lock) {
    atomic_store(lock, 0);
}

void* increment(void* arg) {
    int *id = (int *)arg;
    for (int i = 0; i < 1000000; i++) {
        // acquire_lock(&lock);
        if (i%100000 == 0)
            printf("Thread ID : %d \n",*id) ;
        counter++;
        // release_lock(&lock);
    }
    return NULL;
}
int main() {
    pthread_t t1, t2;
    int T1=1, T2=2;
    pthread_create(&t1, NULL, increment, &T1);
    pthread_create(&t2, NULL, increment, &T2);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    printf("Final counter value: %d\n", counter);
    return 0;
}
```

Insert a screenshot of your output.

```
Thread ID : 1
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 2
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Thread ID : 1
Final counter value: 1973425


...Program finished with exit code 0
Press ENTER to exit console.
```

123.png (7.72 KB)

The correct answer is not displayed for Written Response type questions.

Question 2

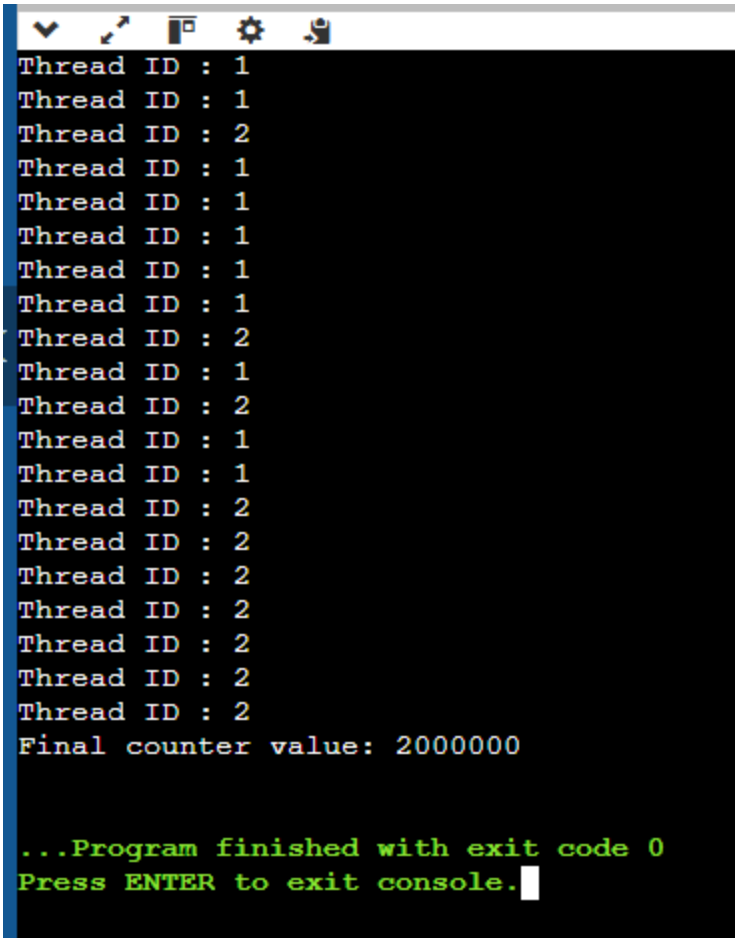
Referring to your results from Question 1, what is the final value of the counter without a race condition? e.g. 32000 ___2000000___ ✔(50 %)

Does the counter value change if you run your code multiple times (Y/N)? ___Y___ ✔(50 %)

Question 3

Referring to your results from Question 1, uncomments lines 21 and 25 and rerun your code,

Insert a screenshot of the output.



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The correct answer is not displayed for Written Response type questions.

Question 4

Referring to your results from Question 3, does the counter value change if you run your code multiple times (Y/N)? ___N___ ✔(50 %)

What is the return value of TSL(lock) if the lock is free? e.g. 1 or 0 ___0___ ✔(50 %)

Question 5

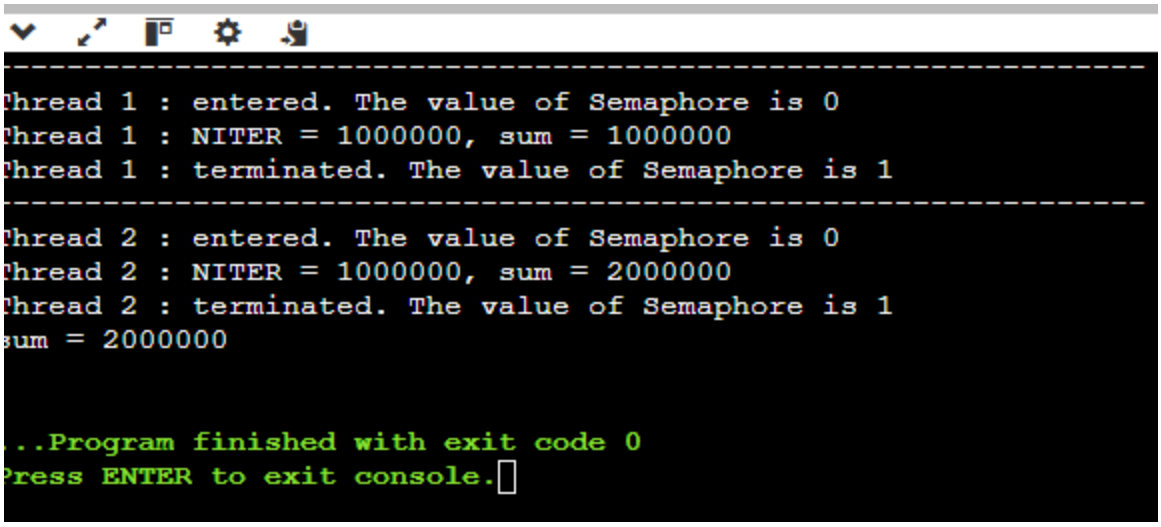
In the following code, the semaphores are employed to prevent the race condition. Compile and run the code and answer the following questions.

gcc sem.c –pthread –o sem.out

sem.c

```c
#include <pthread.h>
#include <semaphore.h>
#include <stdio.h>
#include <stdlib.h>

#define NITER 1000000
sem_t mutex;

int sum = 0;

void * counter(void * a)
{
    int i, tmp;
    int value;
    int *p = (int *)a;

    sem_wait(&mutex);
    sem_getvalue(&mutex, &value);
    printf("------------------------------------------------------------\n");
    printf("Thread %d : entered. The value of Semaphore is %d \n", *p, value);

    for(i = 0; i < NITER; i++)
    {
        tmp = sum;
        tmp = tmp+1;
        sum = tmp;
        if (i==(NITER-1))
            printf("Thread %d : NITER = %d, sum = %d \n", *p, NITER, sum);
    }
    sem_post(&mutex);
    sem_getvalue(&mutex, &value);
    printf("Thread %d : terminated. The value of Semaphore is %d \n", *p, value
}

int main(int argc, char * argv[])
{
    pthread_t tid1, tid2;
    int id1 = 1;
    int id2 = 2;
    sem_init(&mutex, 0, 1);

    pthread_create(&tid1, NULL, counter, &id1);
    pthread_create(&tid2, NULL, counter, &id2);

    pthread_join(tid1, NULL);
    pthread_join(tid2, NULL);

    printf("sum = %d \n", sum);

    pthread_exit(NULL);
}
```

Insert a screenshot of your output.



1111.png (13.06 KB)

The correct answer is not displayed for Written Response type questions.

Question 6

Referring to your results from Question 5, does the sum value change if you run your code multiple times (Y/N)? ___N___ ✔(25 %)

What is the value of the mutex in line 18 (1/0)? ___0___ ✔(25 %)

What is the value of the mutex in line 31 (1/0)? ___1___ ✔(25 %)

What is the sum in line 48 e.g 10000 ? ___2000000___ ✔(25 %)

Question 7

Examine the Java code provided below and utilize the online Java compiler to execute your program.

https://www.onlinegdb.com/online_java_compiler

Main.java

```java
class Monitor {
    private int counter = 0;
    public   void increment1(String name) {
        for(int i = 0; i<100000; i++){
            if( i%20000==0) System.out.println(name);
                counter++;
        }
        System.out.println( name + " : counter = " + counter);
    }
    public   void increment2(String name) {
        for(int i = 0; i<100000; i++){
            if( i%20000==0) System.out.println(name);
            counter++;
        }
        System.out.println( name + " : counter = " + counter);
    }
}
class CounterThread implements Runnable {
    private Monitor mon;
    public CounterThread(Monitor mon) {
        this.mon = mon;
    }
    @Override
    public void run() {
            String threadName = Thread.currentThread().getName();
            if (threadName=="T1"){
                mon.increment1(threadName);
            }else{
                mon.increment2(threadName);
            }
    }
}
public class Main {
    public static void main(String[] args) {
        Monitor mon = new Monitor();
        Thread thread1 = new Thread(new CounterThread(mon));
        Thread thread2 = new Thread(new CounterThread(mon));
        thread1.setName( "T1");
        thread2.setName( "T2");
        thread1.start();
        thread2.start();
    }
}
```

Insert a screenshot of the output.

```
T2
T2
T2
T1
T1
T1
T2
T1
T2
T1
T1 : counter = 194192
T2 : counter = 154608


...Program finished with exit code 0
Press ENTER to exit console.
```

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The correct answer is not displayed for Written Response type questions.

Question 8

Referring to your results from Question 7, have you obtained an accurate counter value after executing your code (hint: Execute your code multiple times.)?

- ✔ ◉ no
- ○ yes

Question 9

Referring to your results from Question 7, implement the specified changes and then run your code again.

```
3   public synchronized  void increment1(String name) {

10  public synchronized  void increment2(String name) {
```

Insert a screenshot of the output.

```
T1
T1
T1
T1
T1
T1 : counter = 100000
T2
T2
T2
T2
T2
T2 : counter = 200000
```

1.png (3.55 KB)

The correct answer is not displayed for Written Response type questions.

Question 10

Referring to your results from Question 9, have you obtained an accurate counter value this time after running your code (hint: do not forget to run your code multiple times.)?

- ✔ (selected) Y, the counter value is 200000
- N, the counter value is less than 200000
- Y, the counter value is 100000

Done